Q1 - General Term and Middle term

The term independent of x in $(1+x)^m\left(1+\frac{1}{x}\right)^n$, is

(1) ${ }^{m+n} C_m$

(2) ${ }^{m+n} C_{n+1}$

(3) ${ }^{m+n} C_{m-n}$

(4) None of these

Q2 - General Term and Middle term

The $(p+2)$ th term from the end in $\left(x-\frac{1}{x}\right)^{2n+1}$,is

(1) $(-1)^p \cdot{ }^{2n+1} C_{2n-\rho} \cdot x^{2p-2n+1}$

(2) $(-1)^p \cdot{ }^{2n+1} C_{2n-p} \cdot x^{2n-2p+1}$

(3) $(-1)^p \cdot{ }^{2n+1} C_p \cdot x^{2p-2n-1}$

(4) None of these

Q3 - General Term and Middle term

The number of terms which are free from radical signs in the expansion of $\left(y^{1/5}+x^{1/10}\right)^{55}$, is

(1) 5

(2) 6

(3) 7

(4) None of the above

Q4 - General Term and Middle term

If in the expansion of $(1+x)^m(1-x)^n$, the coefficients of x and x^2 and 3 and 6 respectively, then m is

(1) 6

(2) 9

(3) 12

(4) 24

Q5 - General Term and Middle term

In the expansion of $\left(5^{1/2}+7^{1/8}\right)^{1024}$, the number of integral terms is

(1) 128

(2) 129

(3) 130

(4) 131

Q6 - Remainder problems

The remainder when 2^{100} is divided by 7, is

(1) 1

(2) 2

(3) 3

(4) 4

Q7 - Remainder problems

The remainder when 5^{99} is divided by 13 , is

(1) 10

(2) 11

(3) 12

(4) 8

Q8 - Remainder problems

The last two digit of the number $(17)^{10}$, is

(1) 49

(2) 48

(3) 47

(4) 46

Q9 - Remainder problems

The last three digits of the number $N=7^{100}-3^{100}$ are

(1) 100

(2) 300

(3) 500

(4) 0

Q10 - Properties of binomial coefficients based on sum

The sum of binomial coefficients of $\left(2x+\frac{1}{x}\right)^n$ is equal to 256 . The constant term in the expansion is

(1) 1120

(2) 2110

(3) 1210

(4) None of these

Q11 - Properties of binomial coefficients based on sum

The sum of the coefficients of all the even powers of x in the expansion of $\left(2x^2-3x+1\right)^{11}$, is

(1) $2\cdot 6^{10}$

(2) $3\cdot 6^{10}$

(3) 6^{11}

(4) None of these

Q12 - Properties of binomial coefficients based on sum

If $\left(1+x-3x^2\right)^{2145}=a_0+a_1x+a_2x^2+\cdots$, then $a_0-a_1+a_2-a_3+\cdots$ ends with

(1) 1

(2) 3

(3) 7

(4) 9

Q13 - Properties of binomial coefficients based on sum

If $(1+x)^n = {}^nC_0 + {}^nC_1 x + {}^nC_2 x^2 + \cdots + {}^nC_n x^n$, then ${}^nC_1 + \frac{2 \cdot {}^nC_2}{{}^nC_1} + \frac{3 \cdot {}^nC_3}{{}^nC_2} + \cdots + n \cdot \frac{{}^nC_n}{{}^nC_{n-1}}$, is

(1) $\frac{n(n-1)}{2}$

(2) $\frac{n(n+2)}{2}$

(3) $\frac{n(n+1)}{2}$

(4) $\frac{(n-1)(n-2)}{2}$

Q14 - Product of binomial coefficients and double summation

The coefficient of x^{49} in the polynomial

$\left(x - \frac{{}^{50}C_1}{{}^{50}C_0}\right)\left(x - 2^2 \cdot \frac{{}^{50}C_2}{{}^{50}C_1}\right)\left(x - 3^2 \cdot \frac{{}^{50}C_3}{{}^{50}C_2}\right) \cdots \left(x - 50^2 \cdot \frac{{}^{50}C_{50}}{{}^{50}C_{49}}\right)$, is

(1) -22100

(2) -11220

(3) 22100

(4) 11220

Q15 - Product of binomial coefficients and double summation

The sum of $\sum_{i=0}^{m} \binom{10}{i}\binom{20}{m-i}$ (where, $\binom{p}{q} = 0$ if $p < q$) is maximum, when m is

(1) 5

(2) 10

(3) 15

(4) 20

Q16 - Product of binomial coefficients and double summation

If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$, then the value of $\sum_{r=0}^{n}\sum_{s=0}^{n} {}^nC_r \cdot {}^nC_s$ is equal to

(1) 2^n

(2) 2^{n+1}

(3) 2^{2n}

(4) 2^{3n}

Q17 - Product of binomial coefficients and double summation

If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$, then the value of $\sum_{0\le r<s\le n}\sum_{r} {}^nC_r \cdot {}^nC_s$ is equal to

(1) $\frac{1}{2}\left[2^{2n} - {}^{2n}C_n\right]$

(2) $\frac{1}{4}\left[2^{2n} - {}^{2n}C_n\right]$

(3) $\frac{1}{2}\left[2^{2n} + {}^{2n}C_n\right]$

(4) $\frac{1}{2}\left[2^{n} - {}^{2n}C_n\right]$

Q18 - Product of binomial coefficients and double summation

If $(1+x)^n = {}^nC_0 + {}^nC_1x + {}^nC_2x^2 + \cdots + {}^nC_nx^n$, then the value of $\sum_{0\le r<s\le n}\left({}^nC_r + {}^nC_s\right)$ is equal to

(1) 2^{2n}

(2) 2^n

(3) $(n+1)2^n$

(4) $n \cdot 2^n$

Q19 - Binomial theorem for any index

The expansion of $\frac{1}{(4-3x)^{1/2}}$ by binomial theorem will be valid, if

(1) $x < 1$

(2) $|x| < 1$

(3) $|x| < \frac{2}{\sqrt{3}}$

(4) $|x| < \frac{4}{3}$

Q20 - Binomial theorem for any index

The general term in the expansion of $(1-3x)^{-1/3}$, is

(1) $\frac{1 \cdot 4 \cdot 7 \cdots(3r-2)}{r!} x^r$
(2) $\frac{1 \cdot 4 \cdot 7 \cdots(3r-2)}{r!} \cdot 3^r \cdot x^r$
(3) $\frac{1 \cdot 4 \cdot 7 \cdots(3r-2) \cdot(-1)^r \cdot x^r}{r!}$
(4) $\frac{1 \cdot 4 \cdot 7 \cdots(3r-2) \cdot(-3)^r \cdot x^r}{r!}$

Q21 - Binomial theorem for any index

The coefficient of x^2 in $(1+3x)^{1/2}(1-2x)^{-1/3}$, is

(1) $\frac{6}{13}$
(2) $\frac{55}{72}$
(3) $\frac{7}{19}$
(4) $\frac{2}{9}$

Q22 - Binomial theorem for any index

The term independent of y in the binomial expansion of $\left(\frac{1}{2} \cdot y^{1/3}+y^{-1/5}\right)^8$, is

(1) sixth
(2) seventh
(3) fifth
(4) None of these

Q23 - Binomial theorem for any index

If $y=\frac{2}{5}+\frac{1 \cdot 3}{2!}\left(\frac{2}{5}\right)^2+\frac{1 \cdot 3 \cdot 5}{3!}\left(\frac{2}{5}\right)^3+\cdots$, then the value of $\left(y^2+2y-4\right)$, is

(1) 1
(2) -1
(3) 0

(4) None of the above

Q24 - Multinomial Theorem

The coefficient of $a^3b^4c^7$ in the expansion of $(ab+bc+ca)^8$, is

(1) 280

(2) 270

(3) 260

(4) 250

Q25 - Multinomial Theorem

The number of terms in the expansion of $(a+b+c+d+e+f)^n$, is

(1) ${}^{n+5}C_n$

(2) ${}^{n+6}C_n$

(3) ${}^{n+7}C_n$

(4) ${}^{n+8}C_n$

Answer Key

Q1 (1)	**Q2** (1)	**Q3** (2)	**Q4** (3)
Q5 (2)	**Q6** (2)	**Q7** (4)	**Q8** (1)
Q9 (4)	**Q10** (1)	**Q11** (2)	**Q12** (2)
Q13 (3)	**Q14** (1)	**Q15** (3)	**Q16** (3)
Q17 (1)	**Q18** (4)	**Q19** (4)	**Q20** (1)
Q21 (2)	**Q22** (1)	**Q23** (3)	**Q24** (1)
Q25 (1)			